1 Planning (15 marks)

Defining the problem (3 marks)

P *r* is the independent variable, *B* is the dependent variable or vary *r* and measure *B*. [1]

P Keep the number of turns on the coil(s) constant. [1]
Do not accept same coil.

P Keep the current in the coil constant. [1]

Methods of data collection (5 marks)

M1 Diagram showing coil and labelled Hall probe positioned in the centre of a coil. Solenoids will not be credited. [1]

M2 Circuit diagram for coil connected to a (d.c.) power supply. [1]

M3 Connect Hall probe to voltmeter/c.r.o.
Allow galvanometer but do not allow ammeter. [1]

M4 Measure diameter or radius with a ruler/vernier callipers. [1]

M5 Method to locate centre of coil.
e.g. determine max V_H; cross rules; projection [1]

Method of analysis (2 marks)

A Plot a graph of *B* against 1/*r* [allow lg *B* against lg *r* or other valid graph] [1]

A Relationship is valid if the graph is a straight line passing through the origin
[if lg-lg then straight line with gradient = –1 (ignore reference to *y*-intercept)] [1]

Safety considerations (1 mark)

S Precaution linked to (large) heating of coil, e.g. switch off when not in use to avoid overheating coil; do not touch coil because it is hot. [1]

Additional detail (4 marks)

D Relevant points might include [4]

1 Use large current/large number of turns to create a large magnetic field.
2 Use of rheostat to keep current constant in coil.
3 Monitor constant current with ammeter to check current is constant.
4 Hall probe at right angles to direction of magnetic field/plane of coil.
5 Reasoned method to keep Hall probe in constant orientation (e.g. use of set square, fix to rule, optical bench or equivalent).
6 *B* is proportional to voltage across Hall probe/calibrate Hall probe in a known magnetic field.
7 Repeat experiment with Hall probe reversed and average.
8 Repeat measurement for *r* or *d* and average.

Do not allow vague computer methods.

[Total: 15]

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